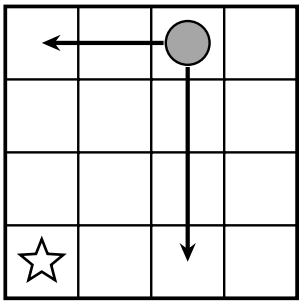
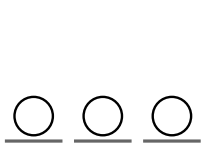


Players take turns. On a board, a move slides the token straight left or straight down, toward the star's side, one or more squares. Whoever moves the token onto the star wins.

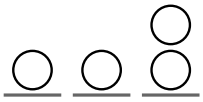
In a game with piles of counters, a move takes one or more counters from one pile. Whoever takes the last counter wins.

Problem 6: A game starts with piles of 52 and 37 counters. Would you rather go first or second? Explain how you can always win, however the other player moves.

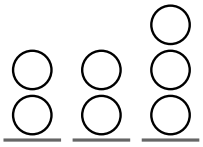
Problem 7: Play many games with three piles from each start. Would you rather go first or second?



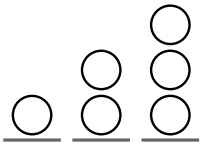
1, 1, 1



1, 1, 2



2, 2, 3



1, 2, 3

Problem 8: Find every start with three piles, each with 1, 2 or 3 counters, where you would rather go second. The order of the piles does not matter. Explain how you know you have found them all.

Problem 9: Find all the starts with three piles, each with 1 to 6 counters, where you would rather go second. The order of the piles does not matter.